The Politic

November 2021
Issue II
The Yale Journal of Politics
and Culture



Waiting for a Haven

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THE FUTURE OF FORESTS



TWO MILLENNIA AGO, a major Chinese river underwent a profound environmental transformation. For thousands of years, the clear waters of the river system that nurtured the earliest seeds of Chinese civilization punctuated vast expanses of northern China. But the translucence of the water soon disappeared. By the fall of the western Han dynasty, China's once crystalline river had become muddied with silt, and the waterway's modern-day name emerged: the Yellow River.

The transformation of the Yellow River was the product of a rapidly expanding empire's deforestation campaign that replaced rich forests with fragile desert. Exposed sediments uncovered in this process saturated the currents, giving the river its characteristic discoloration.

Two thousand years later, the siltation of the Yellow River remains one of the most prominent examples of environmental degradation in Chinese history. The alarming encroachment of desert topography has characterized China's ecological landscape ever since. This encroachment mirrors a global pattern of desertification and deforestation with significant environmental consequences. Today, over 30 million acres of land are lost every year to desertification across the globe, an area over eight times the size of Connecticut. More than 1.5 billion people directly rely on land vulnerable to degradation, and as the climate crisis intensifies, the impacts of this transition to desert land will become more pro-

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Lessons from a Chinese Tree-Planting Campaign

BY **KYRA MCCREERY**

nounced. Any comprehensive climate defense strategy must feature a plan to combat desertification. And interestingly, China has emerged as a global leader in rehabilitating the planet's dwindling forest supply.

In 1978, the Chinese government authorized the creation of the "Great Green Wall," otherwise known as the Three-North Shelterbelt Forest Program. The initiative outlined a plan to reforest China's expanding desert frontier over half a century, representing a sweeping attempt to reverse the nation's spiraling trajectory toward unrelenting environmental disarray.

To date, the initiative has resulted in the planting of over 66 billion trees. Some experts consider the project to be a great success, a shining example of forest reclamation. But the program remains mired in controversy, spurring debates about the extent to which the benefits of afforestation projects outweigh the ecological costs. Are all afforestation projects environmentally beneficial? And to what extent should governments prioritize the planting of trees over other climate policy goals?

UNDERSTANDING THE CONTEXT and implications of China's "Great Green Wall" requires a step back into China's recent past. The latter half of the 20th century witnessed profound changes to the sociopolitical configuration of Chinese society, changes that encompassed nearly every element of China's political structure. In the wake of the Chinese Revolution, the rise of the Communist Party of China (CCP) portended a period of intense industrialization and collectivization, both of which contributed to a sea of short-term environmental crises.

Deforestation soared under the Communist reign of Mao Zedong. A series of industrial reforms known as the "Great Cuttings" sought to enhance China's lumber production to fuel a rapidly socializing economy, but the reforms translated into the loss of nearly a quarter of forest land cover across the country between 1966-1976. The decentralized iron-smelting operations of the Great Leap Forward ravaged forests, and the expansion of deserts during this period worsened the impacts of the famines and food insecurity already so acute during the CCP's reign.

As China recoiled from the devastation wrought by Mao's policies, a series of economic reforms in 1978 enhanced economic freedom and catalyzed modernization following the failures of agricultural nationalization. Under the rule of Deng Xiaoping, China reversed course not only on economic and political issues but also on environmental policy.

The establishment of the Three-North Shelterbelt Forest Program was the most visible manifestation of this newfound ambition to address the intensifying climate crisis. Between 2000 and 2010, China managed to increase its forest cover by 11,500 square miles, an area roughly the size of Massachusetts. Over the next three decades until the year 2050, afforestation efforts are projected to increase forest cover from five percent of the country to fifteen percent.

DESERTIFICATION IS A PERVASIVE PROBLEM in China. "The main problem [China faces] is an oversized population living in the drylands that surpasses the ecological carrying and restoring capacity of this area," said Feng Wang, an associate professor at the Institute of Desertification Studies at the Chinese Academy of Forestry. Feng studies plant behavior and ecohydrology, and his research focuses on creating vegetation models to estimate forest carbon storage. His passion for his work becomes clear as he delves into the relationship between deforestation and desertification impacts. Desertification often results in fierce sandstorms that destroy crops and compromise infrastructure, leaving citizens living on the margins of desert land vulnerable to unpredictable crop failure and instability.

In theory, a project like the Great Green Wall should reduce this instability. If an initiative succeeds at restoring forest cover and sequestering carbon through the strategic planting of trees, it should quell the threat of desertification.

But afforestation is not as simple as planting as many trees as possible. And unfortunately, the Great Green Wall's lack of sophistication could render the entire project fruitless.

Much of the terrain encom-

Comprehensive climate defense plan to combat desertification. A emerged as a global leader in re dwindling forest supply.

passed by the Great Green Wall consists of natural grassland where forests do not naturally grow. Planting trees where they do not belong can often exacerbate existing challenges of aridity. "They're [the Chinese government] going beyond the shelterbelt idea. They're trying to think they can go into grassland and somehow contain the desert by growing trees in places where trees don't grow," explained Robert Mendelsohn, the Edwin Weyerhaeuser Davis Professor of Forest Policy at the Yale School of the Environment, in an interview with *The Politic*. Reinforcing this point, Mendelsohn gestured to two maps he had sent in earlier email correspondence: a brightly colored map of China's topographic variation and an illustration of the expansion of the Great Green Wall into desert. The visual aid highlights the juxtaposition: The new forest is directly overlaid on the topographic regions labeled as desert. Rather than strategically reforest regions amenable to regrowth, the Great Green Wall attempts to force the generation of forests in environments inhospitable to their survival.

Other aspects of the afforestation project have also been called into question. The conservation of already precious water resources in arid regions is a prime concern. Some forest scientists believe the trade-off between the amount of water required to maintain the forests and their environmental payoff is not worth the price. "The question is whether the groundwater in these grassland areas could be put to better use than to grow a row of trees," continued Mendelsohn. "It's just not obvious that this is a smart thing to do."

Another eminent concern is the biodiversity of tree species used to fortify the desert boundary. The Great Green Wall practices the agricultural technique of monoculture planting, or the usage of one species of tree to reforest a degraded territory. A lack of biodiversity can leave an entire forest vulnerable to a single disease, pest, or weather event. Moreover, the monocultural species selected are often not native to the ecological region, introducing a sensitivity to local conditions that might not exist if native species were selected.

Jianchu Xu, a scientist at the World Agroforestry Centre and a professor at the Kunming Institute of Botany, outlined the problems associated with monoculture. This approach to planting provides "little diversity," according to Xu, leaving "almost no habitat for the country's many threatened forest species." Monoculture results in less leaf litter and organic debris than native forests, reducing the range of flora and fauna supported by trees. Xu acknowledges that afforestation projects can offer windbreaks and carbon storage, but these benefits "come at a high cost to other ecological functions."

HOW CAN AFFORESTATION projects optimize environmental conservation? The artificial creation of non-natural forests is unlikely to constitute part of the solution.

According to Mendelsohn, attempting to plant trees in regions of natural grassland is a "waste of resources." A smaller-scale project currently underway in Southwest China, overseen jointly by Conservation International and China's Center for Nature and Society, has attempted to fuse a more diverse medley of plant species ranging from coniferous and broadleaf forests to grasslands and bamboo groves. Through the planting of native species, the project has restored more than 12,000 acres of forest. Other governments in the early stages of reforestation projects might look to this example for a more holistic approach to reforesting a landscape. It seems that the keys to success are to use native species, strategically select environments more receptive to forest ecology, and prioritize water resources in arid regions.

THE CHINESE GOVERNMENT'S failure to make the Great Green Wall truly sustainable presents a puzzle. Why doesn't the CCP seem interested in taking this project more seriously? Is Chinese tree-planting a genuine attempt at ecological sustainability or just a way to camouflage more pervasive environmental devastation?

The answer may lie in a closer examination of China's shifting role on the global stage. In 2013, China adopted the Belt and Road Initiative (BRI), an infrastructure development strategy outlining a plan to invest in over seventy countries and international organizations. Since the inception of the

e strategy must feature a And interestingly, China has ehabilitating the planet's

BRI, China has taken on an active role in economic projects across the globe, often inadequately addressing the environmental fallout of its involvement. In Peshawar, Pakistan, China has invested in the construction of a bus rapid transit system aimed to reduce pollution and resource waste. But in a particularly glaring demonstration of China's environmental hypocrisy, the project required local agencies to cut down thousands of ancient trees and new saplings. Interestingly, the projected end date for the Belt and Road initiative coincides with the optimistic completion of the Great Green Wall — approximately 2050. The question emerges: Has the Great Green Wall always been an illusion?

China is the fastest-growing economy in the world. Annual GDP growth has averaged almost ten percent a year since 1978, the year Deng Xiaoping paved the way for China's modernization and work on the Great Green Wall began. Today, the repercussions of this economic development manifest in China's frontrunner status on the global leaderboard of greenhouse gas emissions.

China has made grand promises to scale back on carbon emissions. At COP26, the U.N.'s climate change conference held in Scotland this November, Chinese leader Xi Jinping urged the world to "accelerate the green transformation." But the Great Green Wall illustrates that China is a land of climate contradictions. Glimmering promises obscure an undercurrent of continued climate apathy hidden beneath the surface. The rapid growth of ecologically-barren forests conceals the remains of trees felled in Pakistan two thousand miles to the west.

THE FAILURES AND LIMITED successes of China's Great Green Wall illustrate the importance of nuanced forest planning for climate recovery. While the dramatic scope of China's afforestation initiative might be compelling, targeted projects that account for the natural environment of the region are far more desirable. Forcing a desert or grassland climate to become a woodland ecosystem ultimately does the planet more harm than good.

It is not too late to restore the forest damage wrought by centuries of land misuse. However, the solution will have to rectify the shortcomings of the existing framework of forest policy. It is not too late to restore the forest damage wrought by centuries of land misuse, but the solution will have to rectify the shortcomings of the existing framework of forest policy. As China's economy exponentially grows, the winding silt-ridden Yellow River continues to be a reminder of the fragile relationship between human inhabitants and their surroundings.

And unfortunately, the Great Green Wall's lack of sophistication could render the entire project fruitless.

BY **KEENAN MILLER**

Math, science, English, social studies, foreign languages, and… computer science? How and why advocates are working to make computer science the newest addition to the American core curriculum.

THE NEW CODE ON THE BLOCK

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Computer Science as a Tool for Equity and Empowerment in K-12 Education

"SO THE BIG TAKEAWAY here is that the more loops we used, the faster I was able to complete the dance, because it was easier for me to understand what the instructions actually were," Aaron Daniels '23 explains to students in a video for Code Haven's Week Three Unplugged Activity. Aaron's simple dance — clapping three times, then moving his hands back and forth to rest on his hips, then his head — was meant to demonstrate how computers think about loop statements.

Volunteers in the Yale undergraduate group Code Haven spend one hour a week visiting classrooms in the New Haven school district to introduce the subject of computer science to elementary and middle schoolers. Their curriculum, developed and refined by Yale students since 2016, works to make learning Scratch — an educational programming language — easy and fun, with activities like dance parties, treasure hunts, card games, and Mad-Libs used in the classroom.

"I always wished I had computer science introduced to me a little bit younger," Eden Gorevoy '24, one of the Code Haven co-presidents, told *The Politic*. "It's important that they get access to it."

According to the 2016 Trends in the State of Computer Science in U.S. K-12 Schools report by Google and Gallup, "the vast majority of parents (84%) and majorities of teachers (71%), principals (66%), and superintendents (65%) say that offering [Computer Science courses] is more important than or just as important as required courses like math, science, history, and English." These are impressive statistics, and an updated report would likely show even higher percentages — but what is the source of this overwhelming belief in the importance of computer science education?

For Leigh Ann DeLyser, the executive director of computer science education equity organization CSforALL, the need for comprehensive, mandatory computer science education, in part, comes from the belief that understanding computers is a form of fundamental literacy. Knowing how to code, the argument goes, is just as important as knowing how to read, write, and perform arithmetic.

"Education is about helping us navigate our world, be productive citizens, [and] be ready for whatever comes next," DeLyser said in an interview with *The Politic*. "Computers influence our daily lives, and understanding the components of the world around you is just as important as knowing that there are cells in your body."

DeLyse is hardly wrong — computers are ubiquitous. Children are more and more frequently spending greater and greater amounts of time on devices, even at very young ages. Although the American Academy of Pediatrics recommends no digital media exposure for children under 18 months of age, and less than one hour of screen time for children between the ages of two and five, over 87% of children in the Upstate KIDS study, which tracks children born between 2008 and 2010, exceeded those limits. Laptops and smartphones are only growing in their importance to the way we live our lives: They undergird our interactions with each other.

Despite this, Jay Lim, who co-teaches Introduction to Computing and Programming at Yale, doesn't believe that a computer science education is necessary for every student yet.

"Still, now, you can get by without really understanding computers," Lim said in an interview with *The Politic*. "And if you really just wanted to live happily, understanding the fundamentals of programming is not really necessary. Maybe 10 to 20 years down the line, our kids will be required to learn programming."

Increasingly, state government leaders seem to agree more with DeLyser than with Lim. Code.org's 2020 State of Computer Science Education report found that 37 states have established academic standards for computer science education, and 20 states require all high schools to offer computer science courses, suggesting that advocacy efforts are having an impact. Asa Hutchinson, the governor of Arkansas and chairman of the National Governors Association, recently announced an initiative to increase access to computer science education nationally. This initiative expanded his efforts in the state of Arkansas, which currently offers computer science coursework in 92% of the state's high schools and is seen as a leader in computer science education. Arkansas also saw significant increases in computer science course enrollment this year.

While, as with any subject taught in schools, not every student will be invested in their computer science education, the goal of these initiatives is to help young people operate in a world increasingly reliant on the structures of computer science.

Code.org writes that "the fields of software, computing, and computer science are plagued by stark underrepresentation by gender, race, ethnicity, geography, and family income." The

front-end impacts of underrepresentation are significant and dangerous — for example, racial biases have been discovered in Google advertisements and AI-driven facial recognition programs.

Though these biases are not embedded into programs intentionally, they cause real-world harm, and the hope is that bringing diversity to the tech industry will help prevent such oversights. Of course, diversifying the tech industry is easier said than done, but the hope is that expanding access to computer science education for students from a broader set of backgrounds will expand the talent pools these companies can draw from and combat the propensity for programming bias.

"There are broad swaths of youth who are being gatekept, who think their path is determined, and I want to show them that's bigger than what they think it is," DeLyser said.

DeLyser knows what it's like to feel like your path is predetermined — that feeling is what drew her to this advocacy work.

"I grew up in rural America, and as a reasonably intelligent young woman…of course I was going to be a math teacher," DeLyser said. Her high school counselor discouraged her choice to even attend college. To pay for her undergraduate education, she held a part-time job writing for database systems on campus. Today, she holds a Ph.D. in Computer Science and Cognitive Psychology from Carnegie Mellon University.

The addition of computer science as a core subject in school curricula is also an exciting opportunity to design and redesign education more broadly with an equity focus.

"Schools should have teachers and admin working together, thinking about challenges facing students, and coming up with practical and worthwhile solutions," DeLyser said.

To encourage this mindset, CSforALL provides what they call the SCRIPT —— which stands for "Strategic CSforALL Resource & Implementation Planning Tool —— to district and school leaders, to encourage the design of computer science curricula with meaningful outcomes and equity in mind. The SCRIPT program takes the form of workshops, rubrics, and other tools for the flexible design of rigorous and realistically implementable approaches to computer science education.

These principles of thoughtful curriculum design are easily cross-applicable to other subjects. For example, DeLyser described an interaction with a charter school leader, who shared with her that they were planning to use the SCRIPT guidelines to reevaluate how all their courses were taught.

Examples demonstrating how individuals can use coding to empower their local communities abound. In March 2020, Avi Schiffman, then a high school student from Washington state, created a highly-trafficked website that collects and displays COVID-19 data. In August 2021, high school sophomore and Tiyya Foundation volunteer Madha Suraj developed an app that connects job seekers with employers, among other things. These young students exemplify how people can use computer science skills to achieve a diverse array of high-impact goals.

"Students with that ability to think about how to tackle problems and how to create a technological intervention to change the world become better problem solvers," DeLyser said. Before stepping into her role as the Executive Director of CSforALL, DeLyser taught computer science at both the high school and university levels. She said she has seen former students use their coding skills to approach critical problems in media, volunteer organization, and health, among other domains. Advocates for making computer science part of the core curriculum in K-12 education also emphasize the opportunities that mastery of coding can open up.

For individuals who aim to enter the workforce, the ability to code is crucial as employer demand for applicants with coding experience has manifested in impressive job prospects. Data from the Bureau of Labor Statistics reveals that computer and mathematical occupations enjoyed a median annual wage of $91,350 in 2020, more than double the overall job market's median annual wage of $41,350. Between 2020 and 2030, the field is also projected to grow at nearly twice the rate of the overall job market. Computer science's promise of good work and financial stability is undeniably appealing to many.

As cyberspace becomes more deeply entrenched in the everyman's life, policymakers have begun to frame the need for a code-literate generation in terms of national security and competition, implying that computer science education in K-12 has national stakes.

For example, the Obama administration's CS For All initiative, signed into law in 2015, set aside billions of dollars in federal funding to expand access to computer science education. Its goal was to ensure "that our nation remains competitive in the global economy and strengthens its cybersecurity."

More recently, in a White House statement from July of this year, the Biden administration wrote that "the cybersecurity threats posed to the systems that control and operate the critical infrastructure on which we all depend are among the most significant and growing issues confronting our Nation."

These concerns are not unfounded. The fact remains that the work available to those entering the American job market increasingly draws upon the field of computer science.

Many students in Lim's introductory computer science course, for example, are not computer science majors. He cited the increasingly interdisciplinary nature of work and research, heightened global interest in machine learning, and expanding fields of data collection and analysis as reasons that students majoring in different subjects

are interested in learning to use computer science tools.

Traditionally, the goal of education is, in part, to help students secure work post-graduation. As computer science becomes more ubiquitous, it follows that computer science education should have a meaningful place in K-12 curricula so that the skills job seekers have will match the skills work providers seek. This will allow American companies to keep pulling from American talent as they grow, and thereby strengthen the economy.

Computer science's importance to national security is also undeniable. By "critical infrastructure," Biden does not only refer to the intangible infrastructure of the internet and all that it supports, although disruptions here could be devastating. He also refers to physical infrastructure — power plants, dams, sewage treatment, and more — that is especially vulnerable to cyberterrorism. In February of 2021, for example, a hacker attempted to fill the water supply in Oldsmar, Florida, with dangerous chemicals. If a plant manager had not caught the attempt in time, about 15,000 customers would have been poisoned. Our physical infrastructure is often not designed with cybersecurity in mind. Both the public and private domains need cybersecurity professionals as concern about vulnerable infrastructure grows.

Computer science also, at least in theory, teaches students transferable skills that allow them to excel in other classroom subjects.

"Computer science is fairly well connected with mathematics," Lim said. "It's all about logic and flow, so by learning to do programming you think about how logic will work, how you can break down some sort of process into little steps and explain each of those steps clearly to a computer."

Aside from logical and structural thinking, it is almost a cliche in the computer science world to emphasize the creativity and perseverance coding can teach. Plus, students like to code — Anna Zhang '24, the other Code Haven co-president, and Gorevoy told *The Politic* that the most satisfying part of Code Haven's advocacy work, for them, was to see students "thrive, get creative, and become connected to everything in their own lives."

All of these skills are valuable, and anecdotal evidence that computer science does something for students is abundant.

"There's some really good anecdotal evidence and stories, and we know that not having [computer science education] is really a blocker for inspiring young youth to pursue careers in the field, especially for disadvantaged students," DeLyser said.

Zhang and Gorevoy agree, emphasizing the importance of showing kids "more people who don't just fit the white male stereotype of what a computer scientist looks like." They also noted that many of the same schools and teachers invite the group to visit annually, suggesting that teachers and students are finding value in learning to code.

Yet it is not yet possible to claim that computer science leads to cognitive benefits or improved outcomes for students via empirical means.

In a recent review, researchers Ronny Scherer, Fazilat Siddiq, and Barbara Sanchez-Scherer called the evidence supporting the cognitive benefits of learning to code "promising." However, the authors ultimately concluded that "more insights into the suc-

"There are broad swaths of youth who are being gatekept, who think their path is determined, and I want to show them that's bigger than what they think it is."

“We need the buy-in of school leaders, teachers, and policy-makers for computer science education to be seen as general literacy.”

cessful transfer [of cognitive skills] are needed to inform educational practice and policy-making about the opportunities to leverage the potential that lies within the teaching of coding.”

DeLyser acknowledged this gap.

“We’re starting to see anecdotally that students who got early exposure to computer science do have better earning potential later in life and are more likely to choose a STEM career,” she said. “But that big data just hasn’t emerged yet because [computer science education] is so new.”

Longitudinal studies tracking long-term student outcomes are in progress. For example, CSNYC, an initiative that built a foundation for computer science education in New York City public schools and that DeLyser formerly served as the Director of Research and Education for, will track students who participated in the program for ten years then publish their results. Still, anything close to conclusive evidence on the matter is several years off.

It’s no wonder, then, that community leaders are somewhat hesitant to invest fully in computer science. Despite increasing implementations of state standards, fewer states — only 20 — require all high schools to offer computer science courses, and only 18 states have developed “timelines, goals, and strategies for providing equitable access for all students, professional learning for teachers, and awareness of the importance of computer science” in communities. Nobody wants to be the guinea pig, especially when a failed experiment can be so costly. But questions regarding the implementation of computer science in classrooms have also developed.

With the benefits of computer science education being uncertain, we are left wondering what making computer science a core subject might come at the cost of. What gives the subject primacy over expanding early language learning programs, or spending more time ensuring that students meet minimum proficiency standards in subjects like mathematics and reading?

At the same time, waiting for conclusive evidence before developing programs has its own costs.

“Giving [K-12 students] exposure as early as possible would be best, but not how universities would teach computer science,” Lim said. “Maybe introducing toys that mimic how coding is done in elementary school would be nice. In middle school, maybe start giving some background knowledge — not a yearlong course specifically for programming, but a basic computer literacy course.” Then, if students enjoy coding, they would be able to take more technical courses in high school.

Zhang and Gorevoy generally agreed with Lim. They envisioned a system focused on exposure, to cultivate a sense of problem-solving and show students fields of study and work in computer science. Computer science would be recurrently visited in the same way many subjects in science are, and students would ideally be able to write a simple script upon graduation.

“We need the buy-in of school leaders, teachers, and policy-makers for computer science education to be seen as general literacy,” DeLyser said. “And we’re getting there… but we need to move beyond that simple ‘yes’ into ‘yes, and [computer science education is] a priority.’”

Implementation depends on whether districts and states have the ability to increase access to computers and high speed internet and prepare educators to teach computer science at a sufficiently high level.

Though challenges in increasing access to quality computer science education abound, preparing youth for a world increasingly reliant upon computers requires that those challenges must be met. Increasing access to this field sooner rather than later, through universal standards developed with equity and empowerment in mind, will only leave Americans better prepared to participate actively in our digital future.

Waiting for a Haven

BY **THOMAS BIRMINGHAM**

Homelessness in Yale's Backyard



PHOTO COURTESY OF ***DESKCT.COM***

Inside Downtown Evening Soup Kitchen's Drop-In and Resource Center, Alvin* was sitting at a small, tucked-away table in front of a booth where a young woman was cheerily handing out coffee. He had laid out a black and gray camouflage backpack sideways and open on the table, and the pattern matched the one on his thin, hooded jacket. As Alvin offered a sturdy handshake, the necklace of red, green, and yellow beads shining against his brown skin was unmistakable.

Despite circumstances, Alvin was in good spirits. Two weeks prior, he had finally received his housing voucher, which will guarantee him some form of shelter for the next year. He thought that should be more than enough time to "get [his] stuff together" and find permanent housing. Crinkles around his brown eyes revealed the smile behind his wrinkled mask.

But for Alvin, good news has been a long time coming. After growing up in Bridgeport, he has faced housing insecurity since the '90s, and by 2008, he was on the street entirely. He slept behind buildings, in parks, in garages, even in the yards of abandoned homes. He said it felt like every door he knocked on just wouldn't open.

"I think when you don't have nothing, you look at people who have," Alvin said. "And you configure in your mind, how do I get there? Because right now, it's food, clothing, and shelter. That's my pie in the sky, that's my picket fence, that's my dream. It seems like it's one of the hardest things to do."

Alvin is just one of the hundreds of people affected by housing insecurity or homelessness in New Haven today.

According to New Haven's most recently available Point-In-Time Count, a systematic manual headcount of the homeless population that happens across the state on one day each January, there were 503 people experiencing housing insecurity that night in 2020. Ward 10 Alder Anna Festa said that when people call the 2-1-1 housing crisis line searching for beds, there can be a waiting list over one hundred people long. Walking through downtown after sundown, one is likely to spot more than one person sleeping on benches on the New Haven Green or nearby sidewalks.

"[Homelessness in New Haven] is dire, very dire," said Karen DuBois-Walton '89, president of Elm City Communities, the housing authority of New Haven.

In the twenty-two months that have passed since that Point-In-Time count, the COVID-19 pandemic has raged, and the number of housing insecure individuals has notably increased even as funding for housing security has not. *The Politic* spoke with city officials, experts, non-profit organizers and even a team of housing program leaders from Utah to better understand the barriers preventing these people like Alvin, this 0.4% of New Haven residents, from finding a home a long time ago.

WHILE ALVIN STRUGGLES to keep a steady roof over his head, at the heart of New Haven, Yale University continues to amass its wealth. In October, the University announced a 40.2% investment return, increasing the endowment by $11 billion in just one year. It now stands at $42.3 billion, 24% of which is not tied to a specific fund and therefore available for general purposes. But the University has historically paid the city of New Haven a voluntary payment of about $13 million — only 0.13% of the $10 billion it has available in general funds.

Right now, Yale has the resources to house New Haven's entire homeless population more than 100 times

over, according to city officials. Right now, this city's most vulnerable population is waiting for something to change.

New Haven Mayor Justin Elicker has called for Yale to pay its fair share since even before his election. One of his biggest campaign goals was to get Yale to increase its giving to $50 million, which has yet to happen. As the city faced a daunting budget crisis over the last year, his demands on the University have increased in intensity.

"I'm calling you to do your part," Elicker said in a press conference last year. "This is the town that you are succeeding in. This is the town that you, Yale University, have an ethical responsibility to contribute to — to ensure that our residents have the resources they need."

LOOKING BEYOND Yale and New Haven, homelessness and housing insecurity have proven to be consistent and dismaying issues in Connecticut's history. According to the Department of Housing and Urban Development, there were 4,316 people facing homelessness in the state in 2010. Eleven years later, there were still nearly 2,600 people experiencing this insecurity the night of the 2021 Point-In-Time count, including over 400 children. Of that total number, 429 were "unsheltered," meaning in just one night of the year, the equivalent of the entire population of Jonathan Edwards residential college was living and sleeping on the streets.

So while homelessness overall has been declining somewhat steadily in recent memory, this data still demonstrates that roughly one in every one thousand people in Connecticut has been homeless throughout the 2000s.

Ward 26 Alder Darryl Brackeen Jr. chairs the Health and Human Service Committee, and therefore oversees approvals of all funding going out to New Haven's Office of Homelessness and other homeless services. Confronted with the current numbers, Brackeen Jr. described himself as a "fierce advocate" for the city's homeless population and said that those in the offices he oversees are doing all they can to make the needs of the homeless known to the larger city government.

"I actually do believe [homelessness] is one of the highest [priorities for the city], probably in the top ten of high priority issues, for sure," Brackeen Jr. said.

New Haven's plan to address homelessness is one page long.

This April, the New Haven Homeless Advisory Commission drew up this five-year plan, spearheaded by the city's Coordinator for Homelessness Velma George, to address homelessness. It would look to "transform" the way the city deals with these issues, with the ultimate goal of creating 550 new housing units by the end of five years. The proposal also suggests creating fifty additional emergency beds, increasing outreach capacity, and creating more "service hubs" in the city to make it easier for someone like Alvin to connect with the right resources.

However, the plan leaves in question the specific funding sources for these efforts. The only concrete measure provided is an allocation of $1.2 million from the city that would be distributed to local homelessness organizations and shelters like Columbus House and New Reach for rapid rehousing, as opposed to permanent housing. The funding for the two permanent housing initiatives — buying out a hotel with 112 units along with the construction of several small, fifty-unit communities throughout the city — remains uncertain.

Coordinator George did not respond to a request for comment on the proposal.

THE PLAN'S ARCHITECTS hope this funding will come from New Haven's new money from the American Rescue Plan. The national relief bill has left the city with a $90 million influx, which has launched a slew of debates as to how to best spend this money. In August, Elicker's office stated at a public event that the funds would be used to specifically target wealth inequality and racial injustice. In September, Elicker's office announced that $12 million of these funds would go towards bolstering the New Haven police, including $3.8 million for new cameras, $3.5 million for a new dispatch system, and nearly $1 million for police overtime.

In New Haven, the idea of homelessness as a "top ten priority" runs counter to the experience of

DuBois-Walton at the housing authority, who said she has yet to hear any word on how this money might go towards housing programs. DuBois-Walton added that while public safety is undoubtedly important, this prioritization of police is a narrow and unsustainable approach to public well-being, indicative of a biased mindset within city government.

"I am concerned because I think too much of our public policy really prioritizes the needs of those who already have wealth," DuBois-Walton said. "We have just decided to invest in a military-industrial complex instead of people. That's the mindset that is guiding public policy."

Bouncing between places in New Haven for years, Alvin sees this "top ten priority" playing out in the city in his day-to-day life. Every day, he witnesses people preparing to spend their nights outside, in the streets. And every day, whenever someone does anything wrong, he said a cop will be there 24/7, 365 days a year.

He wondered why there can't be something that effective for people to get a home or a job.

"If you throw a rock at somebody's car, I bet you fifteen minutes later [the police] are showing up," Alvin said. "But people can walk around for months and months without having a place to live? Hundreds of people. And when you put it in that perspective, it's really, really sad."

Not only does the city allocate comparatively less funding for homelessness, the housing initiatives that the city actually pursues toward this "top ten priority" rarely benefit the homeless and housing insecure population, according to Dubois-Walton. Rather than pushing large-scale affordable housing developments, DuBois-Walton said the city has primarily focused on market-rate developments that many long-time residents would be unable to afford. Last year saw the development of the massive Whit complex in Wooster Square, and the Board of Alders just recently approved another large market-rate development on Long Wharf.

As for affordable housing initiative, DuBois-Walton pointed out that they have not reached the same scale in the Elm City, and said additional cause for concern comes from New Haven's frighteningly low vacancy rate (the percent of housing available to purchase) of 2.7%, as well as the city's placement in the bottom thirty U.S. cities in terms of housing affordability according to a recent study.

"We need to be prioritizing what kind of city we want to be and how we want to have more mixed-income neighborhoods," DuBois-Walton said.

BUT WHILE NEW HAVEN struggles to navigate homelessness and housing insecurity, other U.S. cities are modeling the possibility of productive resolutions.

Despite the barrier of video chat, the orange-red curls framing Salt Lake City's Homeless Engagement and Response Team Project Manager Michelle Hoon shone vibrantly through the screen. She spoke with the clear, measured tone of someone who has addressed these issues for over 15 years. The quiet pride in her voice did not seem unearned, as her home turf in Utah's capital has not only made some of the biggest strides of any city in the nation towards creating effective responses to deal with homelessness, but also cultivated a deep-rooted philosophy that distinguishes those experiencing homelessness from their circumstances.

"Homelessness is legitimately very different than what people perceive it as," Hoon said. "They think that it's like the dude on the corner who is screaming at the sky. But really, it is just an experience that people have. It is a condition that they end up in. It is not a defining feature of their life."

In many ways, the numbers speak for themselves. In 2005, Utah reported over 2,000 people experiencing chronic homelessness, referring to those who have been without shelter for over a year on top of having some sort of "disabling condition," such as a mental or physical disability or substance abuse problems. That same year, Hoon said the state launched an ambitious ten-year plan to end chronic homelessness. By 2015, the number had fallen to well below 200 people, or a 91% drop in the overall rate.

Hoon, as well as Salt Lake City's Homeless Strategies Coordinator Dillon Hase, heavily attributed this success to the statewide adoption of the "housing first" strategy — matching homeless, especially chronically homeless, individuals with a home first and foremost. A place to call their

own. Through this strategy, over 1,100 new permanent supportive housing units, which offer on-site resources for new residents dealing with larger issues, were built within ten years. This approach nearly doubled the number of units in the entire state.

"Ultimately, housing is the solution to homelessness, and we know that," Hase said. "Whatever barriers someone might have, they are going to be better equipped to address those issues once they have the baseline stability of housing. That's why housing first works."

SEVERAL CURRENT CITY LEADERS in New Haven do not buy into this approach. Brackeen Jr., who again directly oversees the way city funds are distributed to New Haven homelessness agencies, said he did not completely subscribe to the housing-first philosophy, emphasizing that growing the city's mental health services will greatly benefit the homeless population.

He also said that in a city of such limited funds, there's only so much that can be done.

"Right now, I'm a strong believer in focusing on immediate relief," Brackeen Jr. said.

Alder Festa, who chairs the City Services Committee, largely echoed Brackeen Jr.'s mentality, saying there are many different reasons why propping up permanent housing first might not be as effective as people think. One common problem she has noticed is that the affordable housing that does pop up is too far from many residents' jobs or schools, racking up potentially unaffordable travel costs. She said if the city spent its money ideally, the focus would be on more emergency beds, mental health programs, and substance abuse prevention programs.

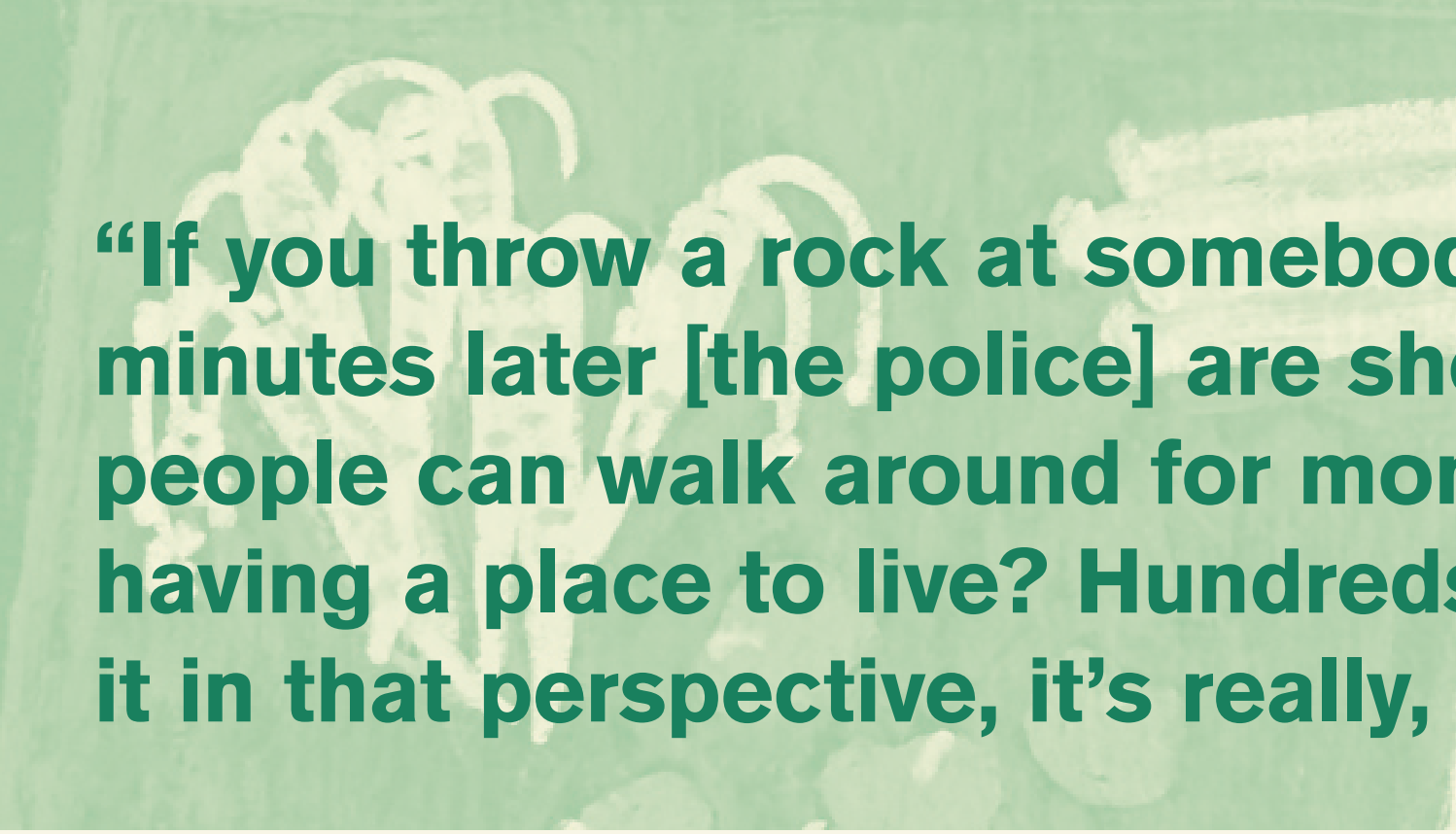
Festa also suggested that there are people in New Haven who simply choose to remain homeless, despite the "amazing resources" offered by the city. She claimed she has noticed a trend of people deciding they want to sleep in the streets. She even expressed doubt that if the city were to "throw money at a homeless person" they would actually use it to try and find housing. For a city like New Haven, she said, there's ultimately no realistic way officials would be able to truly attempt to end homelessness.

"It's no different than a drug addict," Festa said. "I hate to compare it to that. But sometimes they have to reach rock bottom and be almost close to death in order for them to realize 'I have a problem.' 'I'm sick and I need help.'"

Festa neglected to bring up any city data as part of her analysis. On the other hand, both Hoon and Hase were emphatic in stating that across America, the issue of homelessness is highly stigmatized and disproportionately conflated with mental health issues or drug problems. Hase said that 21% of the U.S. population has experienced mental illness, but 21% of Americans are not homeless. He said over 10% of U.S. adults have experience with substance abuse, but 10% of Americans are not homeless.

With this stigma in mind, Alvin recalled being so desperate that

"I am concerned because I think too much of our public policy really prioritizes the needs of those who already have wealth."



“If you throw a rock at somebo minutes later [the police] are sh people can walk around for mo having a place to live? Hundred it in that perspective, it’s really,

he would fake “certain things” like mental illnesses at hospitals in New Haven, Bridgeport, even all the way in Hartford, just on the chance they might be able to offer a place to stay. At the very least, he thought he might be able to get new information about potential housing.

“These people are not homeless because they have an addiction problem or a mental illness,” Hase said. “There are plenty of people that have either one or both of those conditions that are stably housed. They are homeless because they don’t have a house that they can afford. Like, that’s it.”

In an effort to avoid oversimplifying their work, Hoon stressed that Salt Lake City’s housing task forces acknowledge that such factors as mental illness and substance abuse can influence a person’s experience with homelessness and aim to account for these as well. Their housing-first approach does not mean “housing only.”

She did, however, express disappointment that statewide attention has shifted back to short-term, non-permanent housing solutions in light of the increased homeless population thanks to the COVID-19 pandemic.

“In the last few years, we’ve definitely had to shift focus to more of the emergency side of things and less of the housing side of things,” Hoon said. “I think that’s really regrettable actually, because when we have been performing the best is when we had lots and lots of housing units that were coming online.”

INDEED, THE COVID-19 PANDEMIC is just one of many systemic problems that reveal that homelessness extends well outside city lines. From her perspective within the New Haven housing authority, DuBois-Walton said the persistence of homelessness should hardly come as a surprise given wages in Connecticut, and the nation in general, are not keeping up with the housing market.

While she was appreciative of the progress toward a $15 per hour minimum wage in Connecticut, she said this means little considering that housing costs have progressed with rates of inflation, while wages have not. Connecticut’s “housing wage,” or the wage at which most people can pay a typical rent in the state, is currently upwards of $25 an hour. Hoon believes that this gap will only worsen, causing more and more people to become trapped working multiple minimum wage jobs just to keep a roof over their heads.

“I want to see us as a community stop thinking about a minimum wage and start to think about a housing wage,” DuBois-Walton said. “I think we need to start thinking about how we value work and what we think is a reasonable wage to pay.”

However, even when they can afford rent prices, DuBois-Walton said, the woes for low-income laborers do not end. Through her work, she has seen the ways many landlords intentionally discriminate against low-income renters, either due to concern they won’t be able to pay

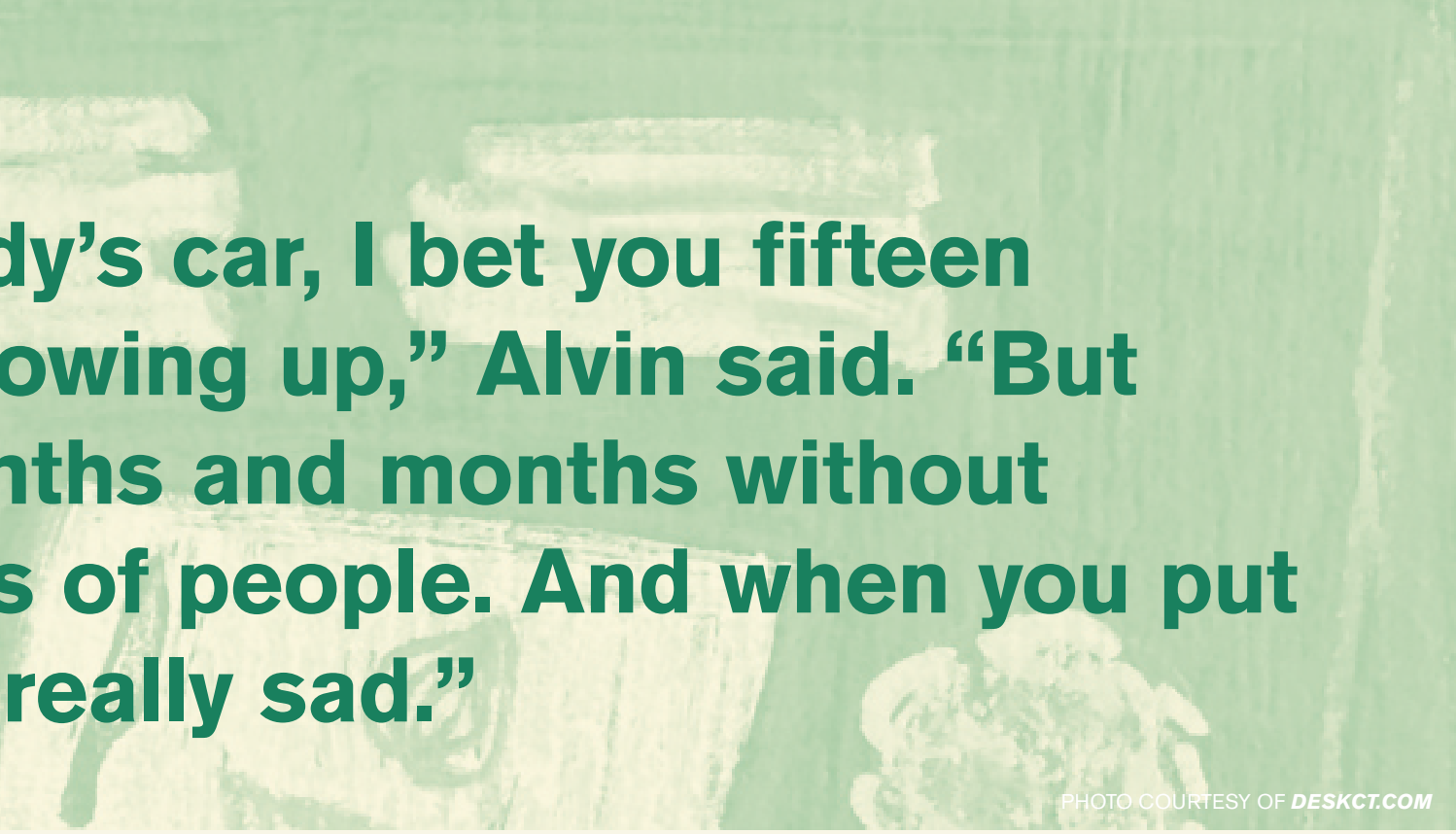


PHOTO COURTESY OF ***DESKCT.COM***

rent, or because people earning more money are therefore able to pay more for an apartment. In light of this, even the federal housing vouchers meant to assist low-income workers with rent worries are insufficient to meet the prices landlords continue to drive up, she said.

Beyond the tangible impacts of this systemic discrimination, Du-Bois-Walton said the leniency accorded to wealth and the scorn to poverty in America would need to fundamentally change in order to address homelessness on a deeper level. She asserted it is this attitude shared by many Americans that facilitates the tax avoidance of billionaire Jeff Bezos and the Nike corporation. It allows a massive institution like Yale to go without paying taxes despite owning over 40% of the real estate in New Haven, while the most vulnerable struggle on its outskirts as victims of a system not designed for their success.

"We assign a lot of personal characteristics and shortcomings on people if they're in poverty as opposed to being able to see the systemic structures that are in place," Du-Bois-Walton said. "We're in a country of tremendous resource, but as a country, we make it very, very hard to be poor."

Alvin remembered the complex and seemingly endless paperwork thrown his way during his efforts to become housing-secure. As ever, obstacles abounded: Alvin was provided incorrect phone numbers and deadline dates as recently as this summer. He emphatically agreed with Du-Bois-Walton that everything seems harder than it needs to be.

Sterling Professor of Sociology and of African American Studies Elijah Anderson, who teaches a course called "Inequality, Race and the City" at Yale, said this kind of rental discrimination disproportionately affects Black residents facing housing insecurity or homelessness. In the state of Connecticut, Black people make up well over 30% of the homeless population, despite only representing around 10% of the population overall. In addition, 30% of all Black people in New Haven are living in poverty, compared to 20% of white people, according to World Population Review.

Anderson highlighted two racialized trends that particularly contribute to the higher rates of Black homelessness he has found in his urban studies work. The first is that as new, wealthier people move into New Haven, white landlords will tend to send current low-income Black residents packing to make room for white residents who can afford to pay more, even if the original residents were successfully paying their rent. The second is that, especially for poor Black residents, it becomes necessary to find a steady job with a workplace that is "receptive to people who look like them," — and many are not.

"You know, it's tough out there," Anderson said. "Poor Black people face persistent discrimination. Anti-Black racism is a serious problem, and so quite naturally you run into problems of housing if you're Black and poor. I

“These people are not homeless because they have an addiction problem or a mental illness,” Hase said. “There are plenty of people that have either one or both of those conditions that are stably housed. They are homeless because they don’t have a house that they can afford. Like, that’s it.”

think a lot of people don’t appreciate that this is what’s happening in the inner city, poor community.”

Because of this, Anderson said the solutions to homelessness will have to be tied to the solutions for systemic racism in this country.

Both Anderson and Du-Bois-Walton, who is also Black, advocated for some form of a universal basic income, or at the very least, a universal housing subsidy. This essentially means that, no matter what, everyone occupying a given community has to get paid above a certain amount annually. They said this could help ensure that even when faced with systemic and racialized obstacles that make it more difficult for Black people to find homes, there is a safety net in place.

“As a nation, we have the capabilities to greatly alleviate human suffering in our cities, and a guaranteed annual income may be one way [to do that],” Anderson said.

For Hoon and Hase, despite the success of their home city, ultimately the discussion about “ending” homelessness in a meaningfully permanent way needs to be a national one.

There is no way someone can travel from Salt Lake City, UT, to Washington, D.C., to New Haven, CT, see all the same trends, and not recognize the homelessness phenomenon as a federal issue, Hoon said. According to the National Alliance to End Homelessness, there are over 580,000 people experiencing homelessness in America today, and Hoon said the only institution with the resources to deal with such a crisis is the federal government. According to Hase, Salt Lake City found that as their homelessness services expanded, more people from nearby municipalities began to flock there in search of relief, so without a nationwide effort, the cities making important progress will continue facing the burden of those that aren’t.

“If it was just a city solution and cities could solve it on their own,

a city would have solved it by now," Hase said.

WITH DEEP-ROOTED NATIONAL PROBLEMS, harshly conflicting mindsets, and a city government that is either unable or unwilling to expand funding for homelessness services, the prospect of even beginning to deal with the issue in New Haven seems at best, daunting, and at worst, totally unattainable. And yet, city leaders continue to ask themselves and their communities what their offices could do if unburdened by the constant and frustrating restriction of funding.

It is by no means impossible to imagine how an influx of funds might arrive in the hands of the city's homelessness services. Thanks to the American Rescue Plan there would be more than enough funds to, for example, send $13 million to the city's offices of homelessness, a sum equal to Yale's current voluntary giving to New Haven. For comparison, the city allocated $1.4 million to homelessness services in the most recent fiscal year. Since the city is spending almost three times that amount this year on police cameras, it is certainly possible to imagine a larger sum of money rerouted towards homelessness, though current prospects are unrealistic.

The other major route to funding is, of course, Yale University. , which announced in November its plans to increase its voluntary contribution by $10 million for the next 5 years. But this increase is still a minute fraction of its total giving capabilities.

Doubling their voluntary payment voluntary payment from this year to send an additional $13 million towards justonly homelessness operations would equate to a $26 million payment or 0.26% of the one-fourth of the endowment reserved for general funds. With an increase in giving to just 1% of this one-fourth, the city could take out the $13 million for homelessness services and still have $87 million the city could spend in other areas. According to multiple city leaders, this funding is not only readily attainable but also its lack is the singular biggest reason why more progress has not been made on this issue.

"I know for a fact that funding has been the biggest hurdle towards implementing most of our plans in their full capacity," Brackeen Jr. said. "Money is definitely our biggest problem."

If $13 million suddenly appeared in the account for homelessness operations, Brackeen Jr. said he would use it to not only fully fund George's five-year plan, but expand it to include even more housing units than currently intended. The plan already would include the construction of more units than homeless residents at the time of the last Point-In-Time count. But with these excess funds, he said the city could expand outreach and resource hub initiatives even further, in addition to being able to fully fund another plan that would repurpose two decommissioned New Haven schools into a system of permanent supportive housing with over 100 beds and a 24-hour staff. All for just 0.13% of one-fourth of Yale's endowment.

Over at Youth Continuum on Grand Avenue, the central outlet for youth homelessness services, Director of Housing Operations Tim Maguire said that with an extra $13 million, homelessness services in the city could make great strides. That figure is nearly three times Youth Continuum's operating budget of roughly $5 million. He said those funds could primarily finance the creation of as many new "deeply affordable" housing units as possible under a "housing-first" model, in addition to dramatically expanding both employment and clinical services.

All for just 0.13% of one-fourth of Yale's endowment.

Were $13 million in additional funds to come under her supervision, DuBois-Walton said she would work to quickly create 1,000 new permanent supportive housing units exclusively for the housing insecure population, which is more than double the number of homeless individuals counted in downtown New Haven in 2020. Actually, she said she could do it for even less: $12 million.

However, her goals were even more ambitious. She said with $200 million, her office would fund the housing subsidy program for every single low-income individual and family in New Haven, meaning all residents would have the money needed to find a home in the city. Every single person. All for just over 2% of one-fourth of Yale's endowment.

"We need funds," DuBois-Walton said. "But it's not that these funds are something we don't have. It's that these funds are connected to our value system. It's about choices now."

WHILE NEW HAVEN DEBATES over less than $2 million, in Salt Lake City, they are currently spending $15 million annually just on their homelessness operations. In cities of comparable size — Salt Lake City houses about 60,000 more people than New Haven's 130,000 — Salt Lake City spends greater than ten times more annually on homelessness than the city where Brackeen Jr. says it's a "top ten priority."

Hoon and Hase attributed successful funding not just to their tax base, but to a coordinated effort with state and local non-profit organizations that ensured their local offices got a large share when the COVID-19 relief bill came along. They both stressed that consistent, sizable funding is such a pivotal element to these issues; continuing to be able to turn to the larger community around them is nothing short of essential.

For New Haven, Hoon and Hase believe that an institution like Yale's willingness to collaborate with its own community will likely be the deciding factor between progress and the continuation of an already dehumanizing crisis.

"[Homelessness] is not a problem cities are really built to solve on

their own," Hoon said. "But there is enough funding out there. If you have the political will and are able to work with the government or non-profit partners you may have, then you will open up much greater funding sources than those from the city itself."

AT ABOUT TEN MINUTES before 5 p.m., back within the noisy and clean walls of Downtown Evening Soup Kitchen's Drop-In Center, Alvin stood up, pushed in the small, lightly-colored chair, and began gathering his belongings into that faded camouflage backpack. As he did so, his voice filled the rapidly emptying space, continuing to speak ardently about the issue that has shaped his life for over twenty years. It was only when he reached the door that he wrapped up and said goodbye with a warm handshake. Then, he left, the cheery woman from behind the coffee stand holding the door for him as he headed out to wait for the bus that would take him to his next meal.

Alvin and the countless others facing homelessness are used to waiting. Waiting for their government to start giving them enough attention. Waiting for the leaders at a place like Yale to wake up one morning and use their unparalleled wealth to help the people sleeping in their backyard. Waiting for the right people to start caring enough to stop all the talk. To do something real.

"I don't just need people to talk to," Alvin said, arms becoming more animated as he spoke. "I got billions of people to talk to. It's not about conversation or communication. I'm trying to get some tangible things going on in my life. So everybody can do the song and dance and a hug and a handshake, but then they won't provide you with the raw materials you need to take it to the next level. And I'm just not dealing with that anymore."



PHOTO COURTESY OF ***DESKCT.COM***

A Matter of Time:

Faced with a Crackdown, Hong Kong Pro-Democracy Protesters Hold Onto Hope

BY **SARAH COOK**

ON THE NIGHT OF JUNE 2, 2020, on his usual walk home, pro-democracy activist Finn Lau was attacked by three people in London. Just two months before, friends in Hong Kong warned him that the Beijing government had placed a $100,000 bounty on his head.

"I tried to stop by a lamppost, but once I stopped, [three people] started to rush to me and kept beating me, kicking me, tossing me, and I thought I lost my right eye immediately because it was so painful," Lau said in an interview with The Politic. "I couldn't see anything. My eyeball was covered with blood. And then I was on the floor. There was a moment I asked myself whether this is the end of my life, and then everything turned into silence."

Lau believes his three attackers acted under orders from the Chinese Communist Party (CCP). He does not believe the attacks were racially motivated, and the three individuals did not try to take his belongings. Additionally, newspapers in Hong Kong exposed his identity a few months later, reporting that he committed crimes of subversion and secession.

The attack on Lau exemplified the CCP's crackdown on pro-democracy activists in Hong Kong, which can largely be attributed to the Hong Kong National Security Law. The law criminalizes acts of secession, subversion, foreign collusion, and terrorism and has been used to justify attacks on pro-democracy groups and activists. Since its enactment on June 30, 2020, the law has allowed the CCP and the Hong Kong government, whose interests are now essentially unified, to justify targeted attacks on journalists and activists and enact election reforms.

On June 4, 2020, thousands of Hong



Kong citizens pushed through barricades, lit candles, and sang pro-democracy songs in Hong Kong's Victoria Park, in an annual vigil held to honor victims of the Tiananmen Square massacre. Three thousand police officers guarded the city to block the gathering under the guise of public health concerns. During the massacre in 1989 — now one of the most heavily censored topics in China — troops stormed in with tanks to a student-led pro-democracy protest calling for freedom of the press, freedom of speech, due process, and accountability.

The CCP's enactment of the national security law later that month caused many groups to drop out of the vigil in 2021, even before local authorities canceled the event due to public health concerns and threatened to subject attendees to five years of imprisonment. Consequently, police successfully barred crowds from entering Victoria Park, and only small groups stood on the park's peripheries. Moments of defiance still punctuated the protests: Activists read a play about Tiananmen Square, searched for pro-democracy postcards, and wrote the numbers indicating the date of the massacre — six and four — on light switches.

"I always think that Hong Kong is a dormant volcano," Lau said. "If you look at the protest this year, during the June 4 vigil, Hong Kongers tried to protest again, even though they tried to block Victoria Park, but still Hong Kong tried to protest on the street. I think we lost momentum. It's just a matter of time another uprising will come."

Months later, on August 23, 2021, the Hong Kong Alliance in Support of Patriotic Democratic Movements of China, organizers who often lead this event, disbanded.

"China's recent crackdown in terms of the national security law and the disbandment of pro-democracy groups, while inevitable in the long-run, is a violation of this agreement and very likely a reaction to the protest movements which have grown in size and frequency since the 1997 handover," Jason wrote in an email to The Politic. Jason is a Hong Kong citizen who asked to remain anonymous.

Some Hong Kongers view China's efforts to exert control over Hong Kong as an unwarranted encroachment on Hong Kong liberties, while others see them as inevitable. Mel Adams '24 grew up in Hong Kong, and her family still lives there.

Adams said her mom, who is a Hong Konger and grew up in Hong Kong, sees what China is doing as inevitable. "At least with respect to my mom, who is a Hong Konger and grew up there, it's like of course China's doing this. It's almost like what the terms and conditions of the [British] handover were," Adams said.

HONG KONG has an extensive history of colonial rule. In 1898, Britain pressured China into signing the Convention for the Extension of Hong Kong Territory, leasing the city for 99 years. In 1941, Japan invaded Hong Kong, but Britain reestablished power in 1946 and remained in power until 1997.

British and Chinese delegations signed the Joint Declaration of 1984 to begin transferring control of Hong Kong. The "one country, two systems" plan, which took effect in 1997, allows Hong Kong to be a special administrative region of China (SAR) until 2047, after which it can keep its own economy and legislative system. The agreement guaranteed Hong Kong its own legal system, rights to freedom of speech and assembly, and multiple political parties for the next 50 years. However, recent events suggest that China has decided to de facto end the terms of the agreement prematurely.





"Unfortunately, Hong Kong has never really had any say over its future. Since the onset of colonial rule, its fate has been determined by outside parties and treaties without local input," Jason said.

The crackdown from China on Hong Kong throughout the last two years began after protests in 2019 over an extradition bill that would allow criminal suspects to be sent to mainland China. Hong Kong Chief Executive Carrie Lam withdrew the bill in October 2019, but the protests continued with additional demands for the Hong Kong government.

"I think our leader, Carrie Lam, was the first one to have caused all this trouble. The extradition treaty was something that everybody was opposed to, she brought it up, and that created the spark," George told The Politic. George is a Hong Kong citizen who asked to be given a pseudonym.

But George believes that the protests eventually became too extreme. When they became violent, the Chinese government reacted swiftly and forcefully; to George, the protesters escalating their demands was not an "educated gamble" since it pitted a relatively small number of protesters against the population of mainland China.

It was shortly after the 2019 protests ended when the standing committee of the National People's Congress of China enacted the national security law, ostensibly a direct result of the protests.

"It was getting to the extreme, so they had to do something so now everyone keeps on going to the extremes. And now, at the end of the day, we sort of ended up losing all our rights, which is sad, but it's evolved that way and there's not much that can be done," George said.

DENISE HO, professor of history at Yale, doesn't think the CCP has merely been reacting to intensified protests. "What you see happening from 2019 onwards [is] a

"I couldn't see anything. My eyeball was covered with blood. And then I was on the floor. There was a moment I asked myself whether this is the end of my life, and then everything turned into silence."

way of using legislation and using interpretation of Hong Kong laws to eliminate opposition to pro-Beijing candidates ad interest in Hong Kong," Ho said in an interview with The Politic.

Ho said there are three main avenues that the CCP can take to exert control over the Hong Kong government. The first is going after pro-democracy activists who could run for office. The second is silencing and prosecuting journalists, and the third is creating an environment where protesters feel unsafe because of the national security law.

The national security law and other electoral reforms in Beijing have effectively kept many pro-democracy activists from political power. Recently, groups have had to disband both due to active threats and arrests to their leaders, as well as fear of future persecution.

The Hong Kong Alliance in Support of Patriotic Democratic Movements of China had been the only group calling for the end of "one-party dictatorship" in China before it dissolved in September. Group members were charged with not providing details about the group's membership and activities to the police by Sept. 7 after authorities requested this information in August. Leaders of the group were charged with subversion under the national security law, and police began arresting the 32-year-old group's leaders in September.

This included Chow Hang-tung, vice-chairwoman of the alliance. She was sitting in her office in Hong Kong's central business district one September morning when she heard police knocking at the door. "Any words of farewell for me?" Chow wrote in a Facebook post while police were ringing her doorbell. She was arrested that day for "publicizing an unauthorized gathering" after telling those planning to attend the June 4 vigil to light candles.

Additionally, the Hong Kong Confederation of Trade Unions (HKCTU) voted to break up on October 3, 2021, citing "political uncertainty." According to pro-Beijing media outlets, the group's activities violated the foreign collusion clause of the Hong Kong national security law, and many group members said they faced threats to their safety. The group also includes the Hong Kong Professional Teachers Union, which voted to disband on September 11 after being condemned in the People's Daily, the newspaper of the CCP. Since the beginning of 2021, 29 unions have disbanded due to fear of facing threats or jail time.

These arrests, along with election reform measures, are also meant to keep pro-democracy candidates from taking office. One such election reform bill was passed by Hong Kong's legislature on May 27, 2021. The new law allows the security department to vet potential candidates and created a new committee to determine if

"Unfortunately, Hong Kong has never really had any say over its future."

candidates are "patriotic." The seats in Hong Kong's legislature also increased from 70 to 90, with 40 of them elected by a pro-Beijing "Election Committee."

Ho said that China traditionally allies with Hong Kong elites, and these election reforms have only bolstered their close relationship. There is essentially no difference between the interests of the CCP and those of Hong Kong's ruling elite, Ho claimed.

"I think that it's very clear now that Hong Kong's political leadership cannot act independently of Beijing. I think this is different from earlier generations of Beijing cultivating ties with Hong Kong elites," Ho said. "Now there's no question that, unless one openly demonstrates their political loyalty to Beijing, they won't be able to participate in Hong Kong's political system."

George thinks differently, arguing that the vetting process allows Hong Kong to choose strong leaders and filter out people who only "spout ideology for the sake of it."

"I know some of the leaders here. And they don't spell communist ideology. They are people who are well educated and they do care for the people," George said.

The Chinese government also seeks to assert control over Hong Kong by systematically silencing and persecuting members of the press. Jimmy Lai, who founded the best-selling Hong Kong tabloid newspaper Apple Daily, was arrested in April for his involvement with protests in 2019, according to the BBC.

"It is our responsibility as journalists to seek justice," Lai wrote. "As long as we are not blinded by unjust temptations, as long as we do not let evil get its way through us, we are fulfilling our responsibility."

Eight other activists were jailed for participating in the same protest as Lai on August 18, 2020, and given a minimum sentence of eight months. Lai was also given an extra two months for involvement in protests on August 31, 2020, according to the South China Morning Post. Lai's arrest led to the closure of the Apple Daily, which had been very critical of the CCP and the Hong Kong government.

LAU IS CURRENTLY STILL IN EXILE in the United Kingdom, and his own experiences of assault and detention in early 2020 have fueled him to fight for democracy in Hong Kong, regardless of the risks associated with the national security law.

"I'm not afraid to face such risks because after being arrested in Hong Kong once and after being in London, my level of determination just elevated to another level," Lau said. "I realize I'm willing to sacrifice my life, so I'm not afraid at all."

But some aren't so optimistic about Hong Kong's democratic future, speculating that it will become like any other major Chinese city.

"The central government is looking to rein in the territory through economic

integration, as seen in the Greater Bay Area plans, social integration, à la flag raising, national education, and national anthems in all schools, and political integration with increasing control over governance in the territory," Jason said.

The Greater Bay Area Plan that Jason referenced would link Hong Kong to cities in China by building financial, infrastructural, and technological ties. With this project, China hopes to mimic Silicon Valley by creating an "innovation cluster." Under the plan, Hong Kong would act as a finance and trade hub, Shenzhen as a tech hub, and Macau as a center for tourism and trade.

The prospect of such a change is not unwelcome for some Hong Kong citizens.

"These changes have probably made life more stable for us, because instead of walking out in the street and seeing protests and wondering what's gonna happen in some ways, breaking down. It's created a level of stability. So, personally, it doesn't really impact me at all. If anything, it's made things more stable," George said.

George expects Hong Kong to move towards a system with an authoritarian government led by meritocratically selected officials, much like the rest of China.

Lau, by contrast, holds out hope for a democratic Hong Kong and other areas in China's power.

"30 or 40 years ago, many people said that the collapse of the Soviet Union was a dream, but in the end, we know the history, and I think the world and Hong Kong could use this strategy, and Xinjiang, Taiwan, and Hong Kong could be free one day," Lau said.

On the other hand, Adams said it still feels like there is still greater tolerance for free speech in Hong Kong than in mainland China, but the threats to protestors still create a sense of fear.

"I definitely think that there's more room, but I don't think there's much room. I wouldn't be afraid for my life if I were to protest in Hong Kong. In China, [protesting] just wouldn't happen," Adams said. "I would be afraid. I guess it's a bit more subtle, but it's also very insidious, the way the Chinese government interferes with Hong

"Since the onset of colonial rule, its fate has been determined by outside parties and treaties without local input."



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LANDS

BY **AXEL DE VERNOU**

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ARTIFICIAL INTELLIGENCE REGULATION IN THE U.S. AND CHINA

FOUR TECTONIC PLATES lie beneath the land of artificial intelligence. They are shifting in the U.S.-China landscape and are positioned for a monumental collision. Each one proposes the most advantageous way to regulate the data, robotics, and electronic systems that have become an integral part of global society. They all owe their origin to the 'AI race' between the U.S. and China.

In one corner is the trial-by-error, principally laissez-faire, corporate American plate, bearing the marks of tense friction between government and private companies. It experiments with different solutions and scales back after harm is detected. In another corner rests the towering Chinese plate, committed to pre-deployment, government curation, and central control. Instead of building on the errors of private sector innovation, the government tests its new technology before widely disseminating it to the public. Europe, trying to get the best of both worlds, is the third tectonic plate. And hovering alongside the three contenders is the fourth plate: industry itself. The private technology sector prioritizes proactive self-regulation and assures the government that it can manage regulation on its own.

"These plates are already starting to hit each other and cause earthquakes," said Professor Nathaniel Raymond, a lecturer at the Jackson Institute for Global Affairs who offered the metaphor in an interview with *The Politic*. "These earthquakes will be

"These earthquakes will be integral to the shape of international trade between nations."

integral to the shape of international trade between nations."

Before framing the development of AI as a competition, which Raymond's analogy suggests, American politicians should seek to understand the specific technological objectives of foreign governments to avoid frantically racing towards an undefined finish line. AI can bring together communities and encourage societal progress, but if left unregulated, companies and users are mutually worse off. The damage caused by foregoing ethical considerations has taken a toll on American companies. Facebook has recently been scrutinized for not addressing harmful body image and misinformation algorithms on its platforms. A few years ago, Google committed a serious blunder when its Photos application labeled Black users as gorillas.

Jin, a Chinese-born student at the University of Southern California who wishes to remain anonymous, found himself in the care of an American foster family at age 10. He spoke to *The Politic*, reflecting on the aftershocks of the U.S.-China collision. He returned to China during the COVID-19 pandemic and stayed there while cases remained low before beginning college in the United States. Jin vividly remembers China's contrasting way of integrating technology into society. A QR code prevented him from entering his hometown following the outbreak of the pandemic. Surveillance cameras at a Chinese tech company he visited could identify someone based on their gait. Face scanners lined the entrances and exits of train stations. "People think that that's how life is supposed to be in China," Jin said, noting that they aren't given an alternative.

These differences are particularly pertinent following China's recent rollout of ethical AI guidelines published by the Ministry of Science and Technology (MOST). The provisions aim for ambitious goals such as promoting justice, safety, and harmony with AI. They emphasize abiding by ethical norms and standards to maintain openness and empower ordinary citizens on digital platforms. To do so, the Chinese government is expanding its oversight by regulating Big Tech companies that traditionally control data and content recommendation. The country's top-down model makes this process much easier than in the United States since Chinese technology companies heavily depend on what the government ordains.

BEFORE DIVING INTO the Chinese model, it is helpful to understand how the four tectonic plates will likely slide in upcoming years. When varying approaches to regulating AI collide, the impact is far-reaching.

In the United States, a clear pattern of harm inflicted on individual users is typically the impetus for AI regulation. For instance, when the American criminal justice system implemented facial recognition technologies for predictive policing, certain racial prejudices — including different levels of policing based on the demographic makeup of certain neighborhoods — were identified after the harm was done. According to Raymond, the U.S. prioritizes market protection, and responses to ethical breaches succeed only if they manage to trigger preexisting legal precedents. In American democracy, the legislature determines the degree of oversight placed on tech companies, resulting in gridlock between conflicting parties.

The U.S. is known to be worryingly slow at responding to ethical concerns because of its lack of a centralized AI policy. The speed at

which emerging technologies are deployed rapidly outpaces ethical and legal oversight. The Collingridge dilemma summarizes this best: It is easiest to regulate a piece of technology early in its development when it is not widespread, but by the time you understand what its problems are, there is no way to reverse its prejudicial effects because it is so entrenched in society. "This has become an excuse for bedeviled technology policy in the last 40 years," said Wendell Wallach, a bioethicist at the forefront of AI ethics policy in China, a nascent field of study.

On the other hand, China is free from these quandaries because the government tracks the effect of its technologies before releasing them, according to Raymond. Through pre-deployment initiatives, the CCP curates data about how citizens respond to new sites or platforms before gradually introducing them to the general public. This is not a strategy exclusive to technology. China goes through a period of loosening control until the central government feels threatened by a particular development, which engenders a strong backlash.

A key example of this was Jack Ma's growing autonomy from the Chinese government as Alibaba, the country's largest e-commerce company, rose to the top of global markets. The founder failed to report the company's acquisitions and transactions to the government, which eventually led to his abrupt disappearance. This is consistent with the justice and safety-oriented provisions proposed by MOST, which emphasize curbing the power of Big Tech companies.

China "reap[s] the advantages of central control until they reap all the disadvantages," said Raymond. "It's like a chess piece that can move ten spaces across the board until it can move none. This is the fundamental paradox of the Chinese government."

The country benefits from pre-deployment and staunch oversight until it limits its innovation. However, the Chinese approach is not inflexible — in fact, it's quite the opposite. The AI ethics guidelines released by China earlier this year outline how "agile governance" and "strategic decision-making" are crucial elements of their AI strategy.

In these guidelines, China has been careful to not explicitly mention its enlargement of governmental influence. The enumerated recommendations omit any detail of what the CCP can or cannot do while targeting companies like Alibaba. As the government takes on a more proactive role in promoting security and justice, it has instilled a feeling of trust amongst most of its citizens. Jin compared his time in the U.S. and China and observed that very few citizens in the latter country harbor any suspicion of encroachment or fear that leaders are overstepping their powers. Faced with much less skepticism than the United States, China can permit itself to launch new mandatory technologies that citizens will almost unilaterally accept.

Jin has experienced both worlds and weighed their respective benefits. "The Chinese media is focused on patriotism" to a degree that encourages the country to "overlook the mismanagement in the government," Jin said. There is only "one narrative about the common good." Jin mentioned the unethically sourced cotton in Xinjiang as an example of a stubborn attempt by the Chinese to portray the country's human rights abuses as nonsensical attacks from foreign powers. The idea of "common good" is highly problematic because it only applies to issues that the government cares about. "I would be able to read articles from a lot of anti-semitic groups" in China, said Jin. People were also "fiercely racist against African groups in the comments of many articles," and would "target you for not being patriotic enough" if you made a false move concerning Taiwan. This should not be seen as a fault in Chinese culture. If anything, it shows that Chinese citizens only have one source to turn to — the government — since they don't look up to industry in the same way Americans do.

Nonetheless, Jin maintains that China's quick recovery from the pandemic demonstrates the utility of state-led technologies. The government works with private companies to develop a QR code for everybody. "I was not able to enter my hometown coming back from the United States," said Jin, holding up his phone to show his code embedded in his phone's WeChat app. China was more successfully able to contain COVID-19 because of technological surveillance.

Another way that China is collecting information about its citizenry is through the country's evolving

social credit system (SCS). One of China's major AI advancements, some Americans frequently misunderstand the system as an overarching database applied to every part of a person's life. Jin notes that the type of credit depends on the industry, but that they all operate under the same premise of constructing a stored record. Jin argues that the system follows "common sense" in many instances. "The Central Chinese bank promotes one [variation] — if you don't pay back the money you borrow, it will reflect in your social credit. You don't want a housing bubble," said Jin.

The same is true of laws. Indicted citizens or those with an unfulfilled financial contract have their information stored in credit files that may lead agencies or private companies to make adverse decisions, according to Jamie Horsley, a Senior Fellow at Yale Law School's Paul Tsai China Center. With a good personal history, for example, one does not have to pay a deposit when booking a hotel, explained Jin. He believes that the SCS encourages people to follow rules and fulfill their societal obligations. Horsley reinforced this point. "The SCS records, shares, and in many cases… discloses information about violations of laws and regulations, court judgments, and legally effective documents," she explained. "It does not record social behavior generally."

It also has practical applications. The "qichacha" system enables people to use an app to determine the credibility of a firm. The app tracks the firm's statistics, including employee social welfare and compliance with contracts, so that stakeholders can decide whether to invest in its services. This type of organization intends to prevent crises like housing bubbles.

To Jin, the benefits outweigh his concerns about government overreach because life flows more freely as a result. "It promotes a better society," said Jin. But the stories he has heard from his family about the perils of dissent returned to his mind as he spoke. "My parents talked to me about my relatives being detained for criticizing the government." One of them called "our supreme leader Xi Jinping a pig on a WeChat post" and another was caught "going to a foreign news website," Jin recalled. Red posters line the streets warning citizens not to get scammed by chat rooms or online casinos, using foreboding messages like "One missed step gives ten years of sorrow."

Horsley stressed that the SCS should not be thought of as generating and applying a score. If a court rules that the purchase of an anti-China book or skeptical online comment violates Chinese law, then the violation is "entered into the SCS, but would not result in any 'score' or other additional consequence not already provided for by law," said Horsley. The principal focus of the SCS is to subtly nudge citizens towards a more law-abiding society by recording and publicizing past transgressions. The government intends to do this by making use of blacklists to track non-compliance and red lists for positive acts, but Horsley explained that the SCS is far from reaching

"Nowadays, the civilian isn't only commingled with the combatant, the civilian is the field of battle… they are the target."

this goal.

While the SCS does not currently make use of advanced surveillance technology, it is not difficult to see how the Chinese government is aiming for harmony with the law. Applying pressure to firms and citizens through records is just one way that it plans on getting there.

THERE ARE TWO DIVERGING WAYS to approach the ethical AI dilemma. The first is through a competitive, realist lens. Raymond espouses this opinion through his opening tectonic plate metaphor.

"Nowadays, the civilian isn't only commingled with the combatant, the civilian is the field of battle… they are the target," he said. "It's not just another form of state competition, it is the migration of state conflict and state military and intelligence into physical and digital infrastructure spaces, [including] healthcare, electricity, transport, and banking."

Yale professor of management and political science Paul Bracken affirmed that "the race is just beginning [but] China hasn't won… I happen to believe they are ahead at the moment." However, he believes that multinational corporations are often an overlooked component of the AI race.

"Apple, BMW, and Walmart want to operate in the U.S., EU, and China, at the same time. BMW, for example, makes 40% of its profits in China," said Bracken. These sprawling companies have thus tried to exercise their influence worldwide to reap the benefits that each region provides.

Bracken has witnessed this trend firsthand. "Recently I've been going to Europe a lot. I'm always at the high-priced restaurants that have sprouted up in Brussels. More and more it looks like K Street in Washington, D.C., the street where the big law firms, lobbyists, 'think tanks,' and business groups are located," said Bracken. "[Lobbyists are] trying to get a favorable regulatory ruling from some part of the EU." In other words, company representatives target whichever geographic location is most favorable to expedite the policy that aligns best with their interests. Brussels, the headquarters of the European Union, has thus attracted the corporate eye since Europe's model promises strong incentives.

General Robert Spalding III, a retired U.S. Air Force brigadier general, does not have such a positive outlook about the AI race. "We are losing. To say we lost, however, would be to say that humanity has ended," Spalding said. "As a Marxist-Leninist system, nihilism is a feature of [Chinese] society. This is not healthy where things like AI-powered weapons are concerned. Without the constraints of morality, there is no limit to the good or harm from AI," he added.

Other academics and AI experts prefer to frame technological development as international cooperation instead of an AI race. "There are a lot of people out there who would like to recreate the Cold War, [pushing] the doctrine that we are in a new economic competition," said

"It's not just another form of state competition, it is the migration of state conflict and state military and intelligence into physical and digital infrastructure spaces, [including] healthcare, electricity, transport and banking..."

Wallach. "Is that really what China is about or is that a narrative that some Americans are promoting?"

The international cooperation argument seeks to equally consider the short and long term ramifications of depicting each country's pursuit of new technologies as a race. Abhishek Gupta, Founder of the Montreal AI Ethics Institute and a board member of Microsoft's Commercial Software Engineering team, believes that co-operation is the most realistic way of framing current circumstances.

"I think the AI race as it's been framed is necessarily a losing scenario for anyone who seeks to participate in it because it forces you to [pursue] paths that seem good in the immediate moment," Gupta said. These paths are laced with elements which "do not yield much in the long run on the international front or on the front of making progress."

Despite the profit that could be garnered from relaying technological breakthroughs to other countries, the status quo tends to maintain a competitive atmosphere, so much so that close allies can even be at odds with one another. British researchers in AI companies refuse to work with the U.S. government, for instance, while Chinese firms do. While the U.S. and the UK have recently fortified their maritime alliance, this is surprisingly irrelevant to the AI "race," where it is every country for itself.

Sasha Brown, who helped form the Ethics and Society team at Google's DeepMind, also underlined the inaccuracy of the "race" metaphor because the finish line looks different for each participant. She enumerated three levels of AI which nation-states aspire to reach. The first is Artificial Narrow Intelligence (ANI), which refers to AI that excels humans for one task, like a phone or self-driving car. The next is Artificial General Intelligence (AGI), namely AI that can perform all human tasks but more efficiently. This second step quickly leads to Artificial Super Intelligence (ASI), alluding to AI which is smarter than humans on every level.

The gap between ANI and AGI is significant. The latter is not necessarily a continuation of the former, and not every country is working toward the same goal. Nation-states that prioritize defense and weapons-based enhancement, such as China, are principally concerned with ANI, but American politicians often ignore this.

Brown believes that the idea of a race is also incompatible with the progress that ethical committees have made in recent years. "People speed up and don't put ethics first," she said. International conferences held between academics from across the globe are collaborative, useful ways to expound theoretical guidelines which then need to be applied concretely, according to Brown.

Gupta emphasized the importance of having a well-intentioned, robust leadership team to address AI concerns. "The Biden administration has brought on board people who have a great track record in terms of addressing issues through a broad socio-technical lens," said Gupta, citing Tim Wu and Alondra Nelson as two influential examples.

Mia Dand, CEO of Lighthouse3, a firm which guides companies with the ethical adoption of AI, focused her work on introducing more voices to the conversation about AI. Dand published the first list of *100 Brilliant Women in AI Ethics* in 2018 and funded the creation of a free online directory that helps recruit more Black and LGBTQ+ innovators in the AI ethics field. She addressed the U.S.' pursuit of ethical AI development in light of Facebook's appearances in court.

"The recent shift towards greater whistleblower protections, labor organizing movement[s], and a pro-democracy administration" is reason for greater optimism, she said. Both ends of the spectrum — an AI race that mirrors the Cold War as opposed to international cooperation — are concerned with interactions between nation states charting their own paths on the road of AI development. But how this AI will be governed is an entirely separate issue.

INDUSTRY HAS SOUGHT TO PROTECT itself from regulation under the guise of a "cult of innovation," according to Wallach. Companies excuse themselves from their sometimes harmful technologies by supporting the narrative that government restrictions would intrude on free-market principles and innovation.

"Tech giants have used their massive funds, army of lobbyists, and complicit rat-



ings-obsessed media to support their 'innovation' myth," Dand said. "Any restrictions on their 'freedom' to engage in unethical and illegal behavior is decried as government overreach or 'socialism,' especially in the United States."

Raymond added that Section 230 of the Communications Decency Act indemnifies most Silicon Valley technology companies from civil legal liability since they are treated as intermediaries republishing online speech. As the Collingridge dilemma

suggests, these tech companies cannot predict the effects of their platforms in advance. "They're like Galapagos tortoises that have grown up without any predators," said Raymond. The evolution of these companies has been gradual and unobstructed, igniting ferocious backlash when the American government attempts to introduce restrictions. In a way, this might suggest that China's authoritarian model is favorable to AI governance since companies are not completely unrestrained. Jack Ma's disappearance reminded emerging Chinese companies that misalignment with the government's standards is costly.

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The goal should be to find an equilibrium between both systems. "We need a balanced approach, which includes solid government regulation that prioritizes public well-being... representation from vulnerable communities that are most at risk from these AI systems, and civilian oversight," said Dand.

Dand expanded on the idea of a narrow, exclusive group of people controlling AI reform. "Lately, the AI discourse has been dominated by big tech and their funded academics, who, along with powerful non-profits, are actively helping reinforce the tech-friendly agenda," said Dand. "It's time to shift this power dynamic and make sure that AI benefits all of humanity, not just the privileged few."

How exactly can this equilibrium be achieved if it feels like governments have already exhausted all possible approaches? The answer is to analyze AI governance in a more nuanced fashion. Raymond offered his version of a five-part analysis which can be instructive for this type of exercise.

The first and second solutions fall under hard and soft governance. The former mostly relies on anti-trust laws to rein in overweening technology companies. Soft government, on the other hand, is mostly driven by industry and civil society. "It doesn't do much," said Raymond, pointing to codes and white papers that fail to meaningfully influence regulation.

The third category, according to Raymond, are the revolutionaries. It's not hard or soft government that will solve the problem, "it's inventing entirely new technological and legal tools like blockchain and data trusts. If we invent a new thing for a new problem then we're fine." While the premise is sound, Raymond doesn't believe that it is viable in today's world because of its logistical complications. Another radical school of thought is refusalism, the fourth approach. It posits that all technology should be shut down.

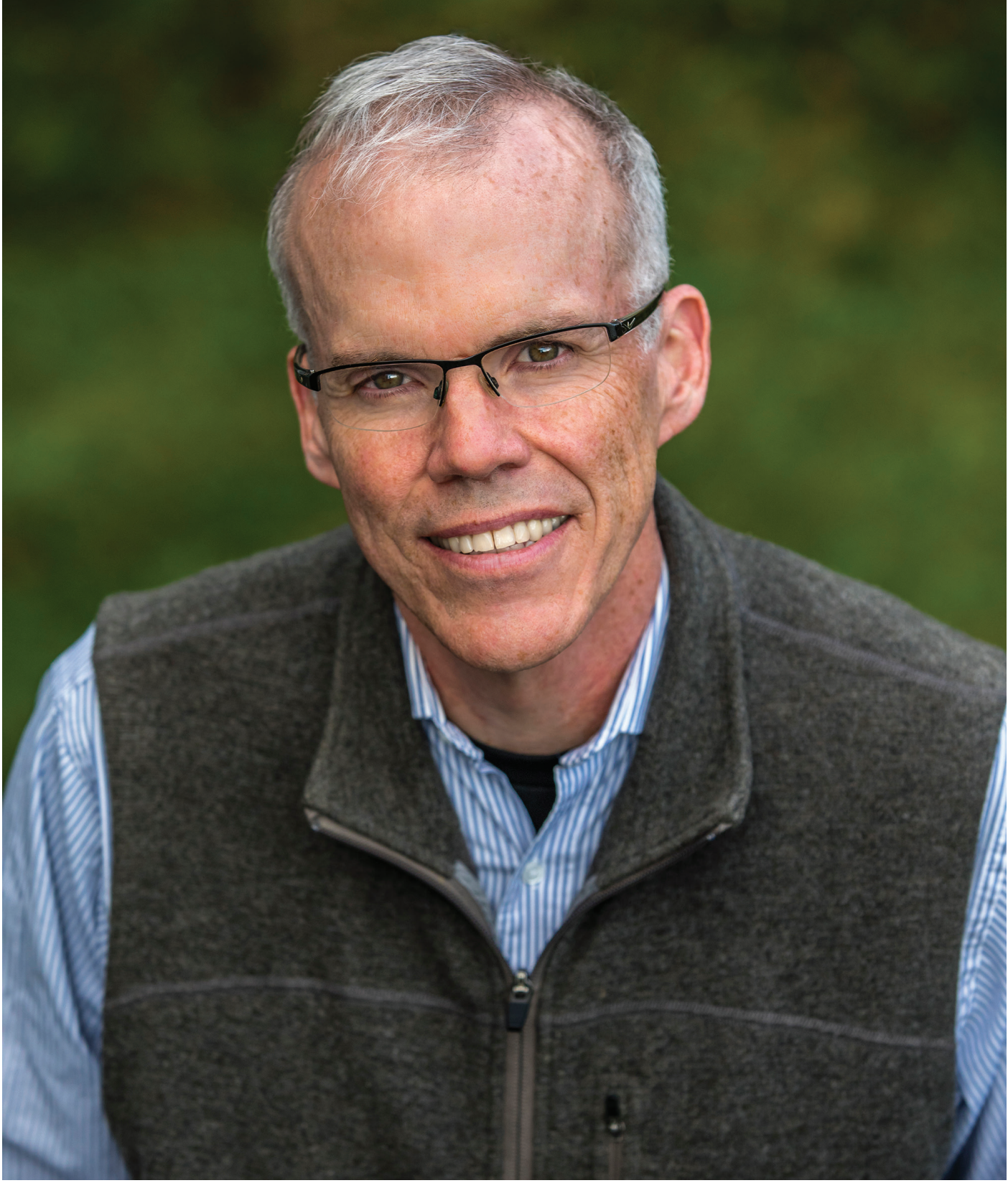
The fifth category, with which Raymond personally identifies, can be called either translationists or neo-normalists. Voicing their general perspective, Raymond said: "We have encountered new technologies before, we'll encounter new technology again. How do we transform existing normative frameworks so that we create new norms that are based on the previous ones? It's not a radical reimagining or an orthodox application. We have to figure out how to translate."

The categories contain more specific approaches. China is typically "very reactive," according to Jin, who attributes many of China's advancements to defensive measures against other threatening countries. China traditionally acts after the effects of a decision have fully materialized instead of taking preemptive steps to circumvent an undesired circumstance, though the same can loosely be said about the United States.

The implementation of ethical AI is multifaceted. Four approaches have been tried: the U.S. and its corporate friction, China's top-down reactive method, Europe's mix of the two, and the shield of industry itself. As these four forces dispute the best way to manage AI, criticism lands on a spectrum: One side maintains that the U.S. and China are in a cut-throat race while the other declares that international cooperation is a more productive way of framing the conversation.

Reconciling these perspectives is perhaps our best option. Yet, the ethical dimension should never be forgotten in considering how the U.S. and China develop their AI. Unregulated technology can be destructive, especially when countries prioritize speed over careful calculation. As with many other international relations issues like climate and human rights legislation, China has created separate standards while the U.S. remains stuck in political and corporate gridlock. Without global standards, AI is an uncontained monster that could go in any direction.

American politicians should recognize the varying objectives that other countries have while placing a firmer grip on companies that have shaped AI debates since their inception. The more concerted and polarized the competition becomes, the less likely that constructive ethical principles will be considered as AI continues to evolve, a dangerous premise with potentially devastating effects on humankind.



Bill McKibben, Photo Credit: Nancie Battaglia

One of the most influential environmentalists of the twenty-first century, Bill McKibben is a best-selling author and the founder of climate group 350.org. His advocacy and journalism have helped shape the global climate movement, including fossil fuel divestment, opposition to the construction of the Keystone XL pipeline, and the People's Climate March. In addition to authoring more than a dozen books, McKibben writes frequently about climate change for publications including the New York Times, New Yorker, *and* Rolling Stone. Foreign Policy *designated him as one of its original 100 most important global thinkers and the* Boston Globe *called him "probably the nation's leading environmentalist." McKibben is the Schumann Distinguished Scholar in Environmental Studies at Middlebury College.*

A Conscience and a Calcu
nscience and a Calculator

A Conversation with Environmentalist Bill McKibben

BY: **BRYSON WIESE**

YOU KICKED OFF THE FOSSIL FUEL DIVESTMENT MOVEMENT IN 2012 WITH AN ESSAY IN *ROLLING STONE* CALLING ON INVESTORS TO SELL SHARES IN COMPANIES WITH HYDROCARBON RESERVES. YOU'VE HAD SOME BIG WINS SINCE THEN. THIS SEPTEMBER, HARVARD ANNOUNCED THAT IT'S DIVESTING ITS $40 BILLION ENDOWMENT FROM FOSSIL FUEL COMPANIES. YALE HAS TAKEN A DIFFERENT APPROACH, RELEASING A SET OF FIVE "ETHICAL INVESTMENT PRINCIPLES," INCLUDING THAT FOSSIL FUEL PRODUCERS MUST MAKE SOME EFFORT TO REDUCE GREENHOUSE GAS EMISSIONS AND SUPPORT GOVERNMENT CLIMATE REGULATION. WHAT DO YOU MAKE OF YALE'S APPROACH?

Yale's just playing games to avoid divesting.

It's really clear at this point that anybody with a conscience and a calculator is divesting from fossil fuels because of the powerful message that it sends. That means Oxford, Cambridge, Harvard, the UC system. The big, huge state pension funds, New York State, New York City, Maine, Quebec. It's by far the biggest corporate campaign of its kind in history.

For a while Yale could hide behind Harvard and pretend that it had some cover. That cover's gone now. It really, really is time for Yale to do the right thing.

We watched this game play out 30 years ago when the subject was divestment from apartheid. Now, as Archbishop Desmond Tutu, who ran the apartheid campaign, has said, climate change is the human rights issue of our time.

It's painful for those of us who really love Yale to see it refusing to take a stand here.

BEYOND YALE, THIS FALL MARKS COP26, THE 26TH UNITED NATIONS CLIMATE CHANGE CONFERENCE. IT HAS BEEN DECADES SINCE THE FIRST ATTEMPTS AT MULTILATERAL SOLUTIONS TO CLIMATE CHANGE. GREENHOUSE GAS EMISSIONS CONTINUE TO RISE. DO YOU STILL HAVE FAITH IN THE CAPACITY OF NATIONAL GOVERNMENTS TO TACKLE THE CLIMATE CRISIS? OR DO YOU SEE OTHER ACTORS, SUCH AS THE PRIVATE SECTOR OR SUBNATIONAL GOVERNMENTS, AS THE MOST PROMISING HUBS OF CLIMATE ACTION?

It's such a huge problem that we need action on every front, unfortunately. National governments have proved incredibly vulnerable to the lobbying power of the fossil fuel industry. It's been excruciatingly slow to see them move into action.

We're finally beginning to see action. In part, that's because movements of people around the world have gotten big enough to counterbalance some of the power of the fossil fuel industry. And partly it's because engineers have done such a good job of lowering the cost of renewable energy that it's less daunting now than it was 10 or 20 years ago.

So, the question is not whether we're going to see movement. We are. The world is inevitably moving towards renewable energy. The question is whether we're going to see it happen quickly enough to even begin to catch up with the physics of climate change.

You know, 50 years from now we're going to run the world on renewable energy. But if it takes 50 years to get there, it's going to be a broken world that we

Anybody with a cons
fossil fuels becaus

run on solar and wind.

Our imperative is to make it happen really fast. The IPCC has told us we need to cut emissions by half by 2030. That's why it's so painful to see institutions like Yale just dragging their feet.

ON THE SUBJECT OF GOVERNMENT-LED CLIMATE ACTION — WHAT DO YOU MOST WANT TO SEE FROM THE BIDEN ADMINISTRATION RIGHT NOW?

Getting this Build Back Better bill through with its climate provisions intact would be the first serious action that the federal government has taken on climate change in all the years that we've faced this crisis.

There are plenty of other things, too — blocking new fossil fuel development, ending subsidies for the fossil fuel industry. But this month, the focus is powerfully on this attempt to get a Build Back Better bill through.

It's worth noting — there are 48 Democratic senators who are willing to support it. That's pretty remarkable. All told, it's the most ambitious piece of legislation since LBJ and the Great Society.

AS A LONG-FORM POLITICAL JOURNAL, WE'RE INTERESTED IN THE POWER OF WRITING. YOU'VE PUBLISHED LONG-FORM OPINION PIECES AT KEY INFLECTION POINTS IN THE CLIMATE MOVEMENT, SUCH AS THE BATTLE OVER THE KEYSTONE XL PIPELINE. TO WHAT EXTENT DO YOU BELIEVE IN LONG-FORM JOURNALISM AND ESSAY-WRITING AS FORCES FOR CHANGE?

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I think they're really important.

You know, I wrote the first book about climate change, which was serialized in the New Yorker back in 1989. Clearly, it was important. But really, there've been lots of important books and articles.

I will say that it took me about 10 years to figure out that we need to do more than write books and articles and have symposiums.

Those are important to win the argument, but it became pretty clear that you could win the argument and still lose the fight — because the fight was less about data and reason than about money and power. A lot of my work in recent years has been organizing.

WHAT'S YOUR NEXT BOOK OR PROJECT?

I have a book coming out in the spring — it's not a memoir exactly, but the subtitle is, "A Graying American Looks Back at His Suburban Boyhood and Wonders What the Hell Happened." It describes growing up in the suburbs of Boston — I grew up in Lexington, MA — and reflecting on American history and race and the kind of mass prosperity that the suburbs represented 50 years ago, and how all those things have curdled in one way or another. The title is *The Flag, The Cross and The Station Wagon*.

The other project that I'm working on is launching this organization we're calling Third Act, which is about organizing people over 60 for progressive action. We've got to get the Boomers and the Silent Generation more in the game. It's been a lot of fun trying to get it off the ground.

If it takes 50 years to get there, it's goin
a broken world that we run on solar an

cience and a calculator is divesting from
se of the powerful message that it sends.

IS THERE A QUESTION OR ISSUE THAT YOU THINK PEOPLE IN THE CLIMATE MOVEMENT AREN'T DISCUSSING ENOUGH RIGHT NOW?

Finally, there's a whole bunch of good climate journalism going on. For the first time in my lifetime — and in the lifetime of this issue — there are now hundreds of talented people thinking about this issue a lot and writing well about it.

If there's a place where we really need more attention, it's on not just the fossil fuel companies, but on the banks, insurance companies, the asset managers that are a financial lifeline. And on their ancillary partners, the groups that support them — law firms, PR firms.

There's a lot of work going on. Law Students for Climate Accountability are trying to get law students not to go to work for firms that are worse than their competitors. Clean Creatives is trying to stop people from heading off to work for PR firms that are just fronts for the fossil fuel industry.

So I think there's going to be lots and lots of places for young people to demonstrate their clear concern for the climate crisis in the years ahead.

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the end of the world is really distressing.

DOES THIS MOMENT THAT YOU'RE DESCRIBING — THE CALLS FOR CLIMATE ACTION RIGHT NOW — FEEL FUNDAMENTALLY DIFFERENT THAN THE PAST DECADES OF YOUR FIGHT?

Yes. We've clearly managed to shift the zeitgeist in powerful ways.

A strong majority of Americans now are concerned about the climate crisis and want action. That number is up 10% over the course of this year.

That reflects the ongoing spate of disasters that we see all the time. It reflects the ongoing organizing that's built enormous movements — especially among young people, junior high and high school students chief among them. It reflects the fact that we have an obvious and easy economical alternative in solar power, wind power, and batteries. Now the thing is just to make it happen fast.

I want to go back to the fact that Yale in particular is managing to really besmirch its name at this point. There's been a lot of great climate science done there, and a lot of great work at the Forestry School, the School of Environment. Remarkable research on these problems. But the fact that Yale is still trying to make a buck off the end of the world is really distressing.

My wife is a proud Yale graduate. The first female Rhodes Scholar in the country. I listen to her when people call her up from Yale asking for money, as they do. She always says, "I would love to be doing that, but not while you're socking it away in Exxon."

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